

Power Query Data Transformation and Cleaning

Dataset Overview

The dataset consists of historical data related to the Football World Cup, with the following files:

- **Matches.csv:** Details of matches played, including team names, scores, and dates.
- **Players.csv:** Information about players, including names, teams, positions, and appearances.
- **Teams.csv:** Details about teams, including team names, groups, and rankings.

Instructions for Data Transformation and Cleaning

1. Matches Table

Transformations:

1. **Promote Headers:**
 - Ensure headers are promoted if they are not already.
2. **Data Types:**
 - Ensure match_id is Whole Number.
 - Ensure date is Date type.
 - Ensure team1 and team2 are Text.
3. **Correcting Errors:**
 - Replace NULL with 0 in the score1 and score2 columns.
 - Remove any rows where date is invalid or missing.
4. **Add Calculated Column: winner**
 - Add a column to indicate the winner (team with higher score).
 - Name the column winner. If score1 > score2, then team1, else if score1 < score2, then team2, else "Draw".
5. **Filter Rows:**
 - Remove rows where team1 or team2 is missing or NA.
6. **Extract Year:**
 - Add a column to extract the year from the date column.
 - Go to Add Column > Date > Year > Year.

2. Players Table

Transformations:

1. **Promote Headers:**
 - Ensure headers are promoted if they are not already.
2. **Data Types:**
 - Ensure player_id is Whole Number.
 - Ensure team is Text.
 - Ensure appearances is Whole Number.
3. **Correcting Errors:**
 - Replace NaN or empty values in appearances with 0.

- Remove any rows where name is missing or NA.
- 4. **Standardize Position Names:**
 - Standardize the names in the position column (e.g., change "strkr" to "Striker").
- 5. **Add Full Name Column:**
 - Combine first_name and last_name into a new column full_name. You need to use a space in between.
- 6. **Filter Rows:**
 - Remove rows where team is missing or NA.

3. Teams Table

Transformations:

1. **Promote Headers:**
 - Ensure headers are promoted if they are not already.
2. **Data Types:**
 - Ensure team_id is Whole Number.
 - Ensure team_name is Text.
 - Ensure group is Text.
 - Ensure ranking is Whole Number.
3. **Correcting Errors:**
 - Replace any NULL values in ranking with -1 to indicate unranked.
 - Standardize group names (e.g., "Grp A" to "Group A").
4. **Add Calculated Column: is_ranked**
 - Add a column to indicate if the team is ranked.
 - Name the column is_ranked. If ranking > 0, then "Yes", else "No".
5. **Filter Rows:**
 - Remove rows where team_name is missing or NA.